

Project Overview:

QUELIFE Animal Shelter Organization's database project is designed to develop a management system for their animal sheltering activities across seven states of USA (Wisconsin, Illinois, Missouri, Montana, California, Texas, and Nevada) to automate their operations. The system will track eight primary entities which are: Facilities, Animals, Adoptions, Adopters, Employees, Sponsors, Donations, and Medical Records, with clearly defined relationships between them. Each entity possesses a set of attributes that attempt to capture the relevant data that is needed including an animal's health status and adoption history, and even an employee's qualifications and a donation's history to a tracking system.

The database incorporates several types of relationships (one-to-one, one-to-many, many-to-many) to capture actual scenarios such as facilities having many animals, adopters being able to adopt many pets, and sponsors donating to many facilities. With Oracle Database and SQL Developer, this system will allow QUELIFE to effectively manage animal records from intake to adoption, streamline the adoption processes, ensure transparent tracking of financial donations, enable better distribution of staff among facilities, and provide full analytical reporting.

The system will be configured for the organization's highest level administrative and operational activities which includes:

- Keeping vaccination and medical treatment records and updating them regularly.
- Documenting adoption cases including reports and follow-up sessions from the adopters.
- Employee information scheduling such as positions, certifications, and assigned buildings.
- Also, for the sake of accountability, managing sponsorship funds by tracking the contributions and outlining the purpose for which the donated funds were utilized.
- Also, reviewing the veterinary care and treatment history in order to improve the speed of providing medical attention.
- With this database, an increase in efficiency of data retrieval, accuracy in record-keeping, and a unified operation throughout QUELIFE's buildings will be achieved.

Entity Sets and Attributes

1. Facility

Facility is a place for housing and caring for animals. The database needs to store information about each facility's location, amenities, and operational details

- Facility ID (Primary Key) – A unique identifier for each facility.
- Facility Name – The name of the shelter facility.
- Location – The state and city where the facility is located.
- Number of Play Areas – Total count of play areas available for animals.
- Health Care Units – Number of medical care units available in the facility.
- Office Space Available – Indicates whether office space for administrative staff exists (Yes/No).
- Temperature Control – Specifies if the facility has heating/air-conditioning (Yes/No).

2. Animal

Animals in the shelter need identification and tracking for medical care, adoption, and shelter management.

- Animal ID (Primary Key) – Unique identifier for each animal.
- Name – The name of the animal (if applicable).
- Species – Type of animal (Dog, Cat, Rabbit, Bird, etc.).
- Breed – Specific breed of the animal.
- Date of Birth – The estimated or actual birthdate of the animal.
- Date of Acquisition – The date when the animal arrived at the shelter.
- Mode of Acquisition – How the animal was received (Rescue, Stray, Owner Surrender, etc.).
- Health Status – General health condition of the animal (Healthy, Sick, Injured, etc.).
- Facility ID (FK) – Foreign Key linking the animal to the facility where it is housed.

3. Adoption

The adoption process records the movement of animals to new homes or people and tracks adoption history.

- Adoption ID (Primary Key) – Unique identifier for each adoption.
- Animal ID (FK) – Foreign Key linking to the adopted animal.
- Adopter ID (FK) – Foreign Key linking to the adopter.
- Adoption Date – The date when the adoption was completed.
- Adoption Fee – Fee paid by the adopter (if any).
- Follow-up Status – Indicates whether a follow-up check was conducted (Pending, Completed, Not Required).

4. Adopter

Individuals or organization who adopt animals must have their contact and eligibility information stored.

- Adopter ID (Primary Key) – Unique identifier for each adopter.
- Name – The full name of the adopter.
- Address – Residential address of the adopter.
- Phone – Contact number.
- Email – Email address of the adopter.
- Previous Adoptions – Number of animals adopted by the adopter in the past.
- Approval Status – Whether the adopter is approved for adoption (Pending, Approved, Rejected).

5. Employee Shelter

Employees handle animal care, adoptions, and administrative duties.

- Employee ID (Primary Key) – Unique identifier for each employee.
- Name – Full name of the employee.
- Role – Job title or position (Veterinarian, Caretaker, Manager, etc.).
- Facility ID (FK) – Foreign Key linking the employee to their assigned facility.
- Education – Academic qualifications.
- Certification – List of certifications related to animal care or shelter management.
- Criminal Record – Boolean (Yes/No) indicating any prior criminal records.
- Contact Info – Employee's phone number or email.

6. Sponsor

Financial sponsors provide donations and financial support for shelter operations and need.

- Sponsor ID (Primary Key) – Unique identifier for each sponsor.
- Name – Sponsor's full name or company name.
- Organization Name – If the sponsor is part of a business, their company name.
- Contact Info – Phone number and/or email address.
- Donation Amount – The total amount donated by the sponsor.
- Sponsorship Type – Type of sponsorship (One-time, Recurring).

7. Donation

Each financial contribution to the shelter is recorded and categorized for auditing and to make the process transparent.

- Donation ID (Primary Key) – Unique identifier for each donation.
- Sponsor ID (FK) – Foreign Key linking the donation to the sponsor.
- Facility ID (FK) – Foreign Key linking the donation to the receiving facility.
- Amount – The donation amount.

- Donation Date – Date when the donation was received.
- Purpose – Purpose of the donation (Medical, Food, Renovation, General Fund).

8. Medical Record

A log of medical checkups, treatments, and health history for each animal.

- Record ID (Primary Key) – Unique identifier for each medical record.
- Animal ID (FK) – Foreign Key linking the record to the specific animal.
- Checkup Date – The date of the medical checkup.
- Diagnosis – Medical condition or diagnosis.
- Treatment – Description of treatment provided.
- Vet Name – Name of the veterinarian or medical professional.
- Next Appointment – The scheduled date for the next medical checkup.

Relationships

1. Facility houses Animals (One-to-Many)

- o Each shelter facility provides housing for multiple animals.
- o However, an animal can only be assigned to one facility at a time.

2. Animal is Adopted by Adopter (One-to-One)

- o Once an animal is adopted, it is no longer available for adoption.
- o Each adoption record connects one adopter to one adopted animal.

3. Employee Works in Facility (Many-to-One)

- o A facility has multiple employees, including caretakers, vets, and managers.
- o Each employee works at only one facility.

4. Adopter Completes Adoption (One-to-Many)

- o An adopter can adopt multiple animals over time.
- o Each adoption record connects one animal to one adopter.

5. Sponsor Funds Facility (Many-to-Many)

- o Sponsors can donate to one or more shelter locations.
- o A single facility can receive financial support from multiple sponsors.

6. Facility Receives Donations (One-to-Many)

- o A facility can receive multiple donations over time.
- o Each donation is assigned to only one facility.

7. Animal Has Medical Record (One-to-One)

- o Each animal has a unique medical history recorded in a Medical Record.
- o A medical record is linked to a single animal.

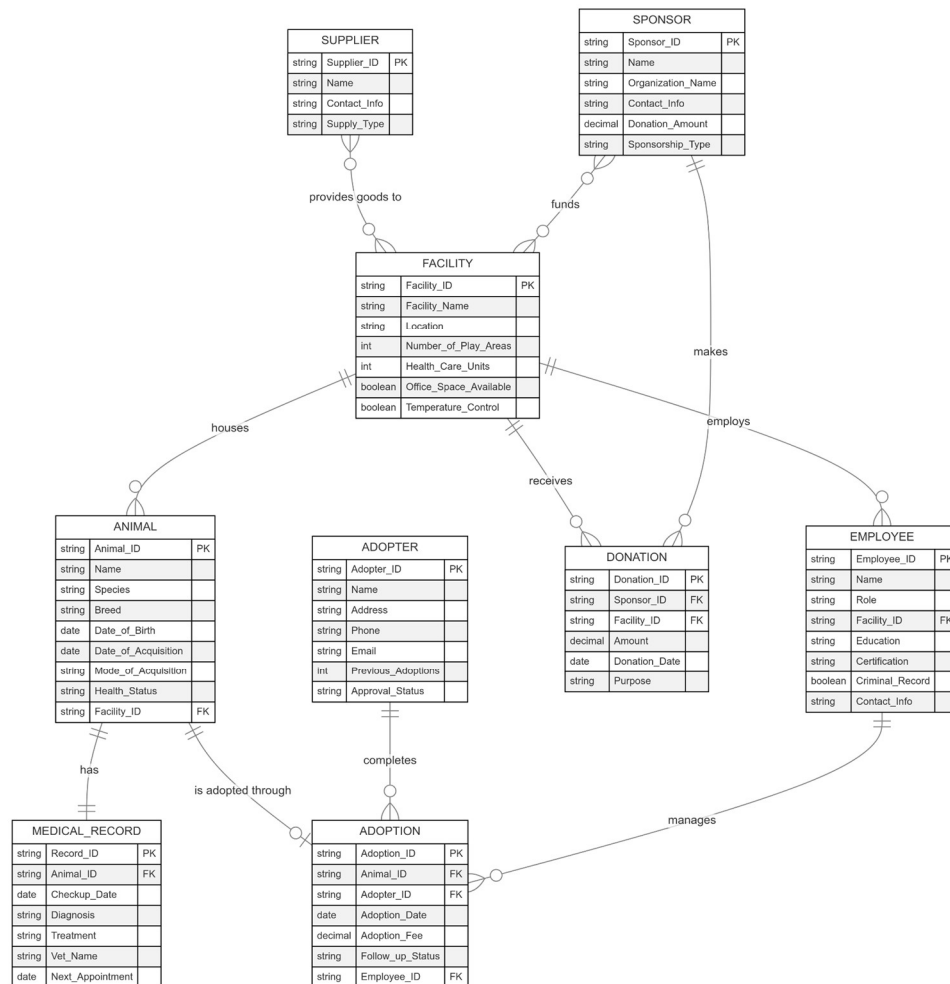
8. Supplier Provides Goods (Many-to-Many)

- o A supplier may provide food, medical supplies, or other materials to multiple facilities.
- o A facility can receive goods from multiple suppliers.

9. Employee Manages Adoption (One-to-Many)

- o Each adoption process is overseen by a shelter employee (e.g., adoption coordinator or shelter manager).
- o An employee can manage multiple adoptions, but each adoption is handled by only one employee

ER Diagram



QUELIFE Animal Shelter Organization has an Entity-Relationship diagram which contains all nine entities with their attributes, primary key, and foreign key fully labeled and structured. The diagram demonstrates all nine entities' relationships which includes one-to-one, one-to-many and many-to-many relationships with their cardinality symbols. It is drawn so that the number of crossing lines is kept to a minimum, and the labels that describe the relationships between entities are clear. This diagram is a good design for executing the database schema in Oracle because it depicts the organizational operational structure spread over seven states.

Individual Contribution:

I have identified the primary and foreign keys and the cardinality of relationships.

Recommended normalization and data duplication that was seen in the database.

Corrected the documents that were incorrect and unapologetically reported all.

Gave further details about the database constraints and integrity.